6 b output.sci (C:\Users\LENOVO\Documents\6 b output.sci) - SciNotes

*6 b output.sci

```
// Test Case: Autocorrelation of a Rectangular Pulse Sequence
// Define parameters
N = 20; // Number of samples
x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5
// Compute the autocorrelation
R_xx = correl(x_n, x_n);
// Create a lag vector for plotting
lags = -N+1:N-1;
// Plot the input sequence
clf; // Clear the current figure
subplot(2,1,1);
bar(0:N-1, x_n, 'b');
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Rectangular Pulse Sequence');

// Plot the autocorrelation sequence
subplot(2,1,2);
bar(lags, R_xx, 'r');
xlabel('Lag');
ylabel('Autocorrelation');
title('Autocorrelation of the Rectangular Pulse Sequence');

```

Graphic window number 0

File Tools Edit ?

